

Probability 1 is Bad Form

Omar

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Rationality Freiburg

Who here has ever said they were **100% sure** of something?

(Be honest — keep your hand up if it was in the last 6 months.)

This is bad form.

Not a crime. **Bad form.**

Like putting your shoes up on the seat in the train.

A pet peeve of mine. I heard it twice last month, from two different rationalists.

Two reasons it's wrong

When you mean it *literally*:

1. You'd have to take **extreme** bets.
2. You can't even **get** there.

...then: "but I don't mean it literally" — and what to say instead.

Reason 1 — You'd have to take
the bet

"I am **100% sure** that Friedrich Merz was Chancellor of Germany on 1 June 2026."

About as solid as a fact gets. Would you endorse this statement?

Then take this bet

If you're right

+1 €

If you're wrong

everything

house · savings · the dog

You said 100%, so there's no downside. Free money — you should be ripping my arm off.

For the mathematicians

$$EV = \underbrace{P(\text{win}) \cdot (+1\text{€})}_{\text{you keep this}} - \underbrace{P(\text{lose}) \cdot \text{everything}}_{\text{this vanishes}}$$

If $P(\text{win}) = 1$, then $P(\text{lose}) = 0$ and the second term is gone.

EV = +1€, guaranteed — no matter how big “everything” is.

Probability **exactly** 1 means: accept **any** payout against **any** stake.

You won't take it?

Then you're not at 100%. We're just **haggling over the decimals.**

Reason 2 — You can't even get there

Cromwell's rule

Oliver Cromwell, to the Church of Scotland, 1650 — *a 17th-century military dictator, on epistemics:*

"I beseech you, in the bowels of Christ, think it possible that you may be mistaken."

Dennis Lindley named a rule after it: never use a prior of 0 or 1.

Why? The math is merciless

Bayes' theorem, in standard form:

$$P(H \mid e) = \frac{P(e \mid H) \cdot P(H)}{P(e \mid H) \cdot P(H) + P(e \mid \neg H) \cdot P(\neg H)}$$

But if $P(H) = 0$, then $P(H \mid e) = 0$ — no matter the evidence. And if $P(H) = 1$, the same holds for its negation.

Start at 0 or 1, and **no evidence can ever move you.**

A concrete example

H = "the coin is two-headed". You're **certain**: $P(H) = 1$.

Now you flip it and see **tails** — that's e , strong counterevidence:

- $P(e \mid H) = 0.001$ (a two-headed coin "should" never land tails)
- $P(e \mid \neg H) = 0.999$ (a normal coin easily does)

Plug in $P(H) = 1$, so $P(\neg H) = 0$:

$$P(H \mid e) = \frac{0.001 \cdot 1}{0.001 \cdot 1 + 0.999 \cdot 0} = \frac{0.001}{0.001} = 1$$

The tails get **explained away**. You stay at 100% — forever.

Probability 1 isn't confidence.

It's: "no observation, ever, could change my mind."

That's not confidence. That's a **religion**.

“But surely $2 + 2 = 4$ is certain?”

No. Even that. The uncertainty isn't in the **math** — it's in **you**, the fallible meat computer running it.

Never made an arithmetic slip? Never trusted a proof with a hole on page 4? Not dreaming right now?

The sliver you keep back isn't for “maybe $2 + 2 = 5$ ”. It's for **“maybe I'm wrong about being right.”**

But I don't mean it literally

"I obviously mean 99.9999%"

Fine for poetry: "I'm 100% behind you", "I 100% love this restaurant". *Knock yourselves out.*

But for an **epistemic** claim — about what is **true** — drop it entirely. Never. Not even as shorthand.

Two reasons, even symbolically

It launders the error. Careful people saying “100%” make it sound normal

Two reasons, even symbolically

cover for the next person who means it literally.

It throws away the information. 99.9999% vs 99%

Two reasons, even symbolically

"wrong 1 in a million" vs "1 in 100", a factor of 10,000. Rounding to "100" erases exactly the part worth saying.

A tiny daily discipline.

Epistemic hygiene. **Rationalist flossing.**

What to say instead

A real number says something

99% → "wrong about 1 in 100"

90% → "wrong about 1 in 10"

Not weakness — a **measurement**. It tells me how much to lean on what you said.

...but no false precision

Say "99.73%" and you'd better defend the **.73**.

Otherwise it's just 100%'s arrogant cousin — certainty in a lab coat.

The goal isn't **more** decimals. It's **honest** ones.

The number is a **gift.**

It tells people how much to trust you — and lets them keep score. That loop is how you get calibrated.

Never 100%. Never 0.

Not because you're unsure of everything

Never 100%. Never 0.

because you're **honest** about being a fallible human running on three pounds of warm meat.

Give people your real number. It's a gift.

Am I 100% sure of everything I just told you?

...No. Call it **99.9%.**

Thank you.